

Companion Results Appendix: GGUF Throughput Prediction

Protocol-complete 3-host evidence for the ICASSP manuscript

Status. This is a locally generated companion appendix. It does not claim publication, archival acceptance, or independent replication.

1 Scope and cohort

The source manifest contains 23 GGUF files. Across 3 hosts (MacBook Pro M4 Max, Mac Studio M4 Max, and RTX 5080), the scored data contain 22 unique files and 53 host–file configurations (42 training and 11 held out). Each configuration has one decode and one prefill observation at each of the 3 measured context depths, for 159 decode and 159 prefill rows. The final protocol selector retains 354 observations from 23 unique files in total: 318 scored rows and 36 reference-probe rows excluded from prediction fits and scores. Probe exclusion is host-specific, so a file used as a probe on one host can be scored on another. All exported rows carry the declared protocol fields. The retry decision, however, uses the worst within-run *decode* CV in an invocation. Prefill rows inherit that invocation’s protocol metadata but are not gated on their own CV. This distinction matters on RTX 5080, where the noisiest prefill row has 39.19% CV even though the associated decode-gate CV is 1.35%.

Host	Backend	Train cfg.	Test cfg.	Scored decode	Scored prefill
MacBook Pro M4 Max	BLAS,MTL	15	4	57	57
Mac Studio M4 Max	MTL,BLAS	15	5	60	60
RTX 5080	CUDA	12	2	42	42

Host	Retried cells	Gate CV med.	Gate CV max	PP CV med.	PP CV max	PP > 3%
MacBook Pro M4 Max	3	1.43	3.04	1.41	4.84	7/57
Mac Studio M4 Max	1	0.88	2.30	0.14	0.96	0/60
RTX 5080	3	0.88	1.65	3.56	39.19	29/42

Attempted configurations with no successful row are MacBook Pro M4 Max: `gpt-oss-120b-MXFP4.gguf`. On the MacBook Pro M4 Max, two raw attempts of the one `gpt-oss-120B` configuration report failed to decode prompt batch, `res=-3`; these are prompt-batch failures, not demonstrated model-load or out-of-memory failures. Host-specific calibration probes are MacBook Pro M4 Max: `Qwen3.8-27B-Q8_0.gguf`, `Qwen3.5-9B-Q8_0.gguf`, `Qwen3.5-4B-Q8_0.gguf`; Mac Studio M4 Max: `Qwen3.8-27B-Q8_0.gguf`, `Qwen3.5-9B-Q8_0.gguf`, `Qwen3.5-4B-Q8_0.gguf`; RTX 5080: none.

1.1 Cohort accounting

Raw records, superseded legacy rows, repeated protocol attempts, selected observations, and scoring exclusions are separated below. A protocol duplicate is a successful protocol row removed by the atomic host–model–phase–depth selector; it is not an additional scored observation.

Host	Raw	Failed	Legacy ok	Protocol ok	Dup. removed	Selected	Probes	Scored
MacBook Pro M4 Max	278	2	132	144	12	132	18	114
Mac Studio M4 Max	138	0	0	138	0	138	18	120
RTX 5080	84	0	0	84	0	84	0	84

1.2 Excluded reference-probe observations

These rows complete the 354-observation selected cohort and make the descriptive matched-host and quantization comparisons reproducible. They are reported only as measurements, never as prediction errors.

Table 1: Selected protocol-complete reference-probe observations excluded from prediction fitting and scoring. Measured throughput and SD are in tokens/s; IDs resolve through Table 2.

Host	ID	Phase	Depth	Measured	SD	CV %	Attempts
MacBook Pro M4 Max	M10	decode	0	77.275	0.201	0.260	1
MacBook Pro M4 Max	M10	decode	4,096	74.687	0.153	0.204	1
MacBook Pro M4 Max	M10	decode	16,384	67.270	0.341	0.507	1
MacBook Pro M4 Max	M10	prefill	0	1553.086	1.682	0.108	1
MacBook Pro M4 Max	M10	prefill	4,096	1416.098	1.799	0.127	1
MacBook Pro M4 Max	M10	prefill	16,384	992.403	10.398	1.048	1
MacBook Pro M4 Max	M12	decode	0	46.499	0.380	0.818	2
MacBook Pro M4 Max	M12	decode	4,096	45.340	0.169	0.372	2
MacBook Pro M4 Max	M12	decode	16,384	42.119	0.938	2.226	2
MacBook Pro M4 Max	M12	prefill	0	865.397	0.661	0.076	2
MacBook Pro M4 Max	M12	prefill	4,096	784.470	4.299	0.548	2
MacBook Pro M4 Max	M12	prefill	16,384	560.769	14.295	2.549	2
MacBook Pro M4 Max	M14	decode	0	14.902	0.054	0.365	1
MacBook Pro M4 Max	M14	decode	4,096	14.433	0.017	0.118	1
MacBook Pro M4 Max	M14	decode	16,384	13.575	0.092	0.680	1
MacBook Pro M4 Max	M14	prefill	0	238.511	7.257	3.043	1
MacBook Pro M4 Max	M14	prefill	4,096	201.115	2.027	1.008	1
MacBook Pro M4 Max	M14	prefill	16,384	161.599	1.227	0.759	1
Mac Studio M4 Max	M10	decode	0	78.751	0.546	0.693	1
Mac Studio M4 Max	M10	decode	4,096	76.142	0.100	0.131	1
Mac Studio M4 Max	M10	decode	16,384	70.660	0.398	0.563	1
Mac Studio M4 Max	M10	prefill	0	1567.538	4.181	0.267	1
Mac Studio M4 Max	M10	prefill	4,096	1430.704	3.482	0.243	1
Mac Studio M4 Max	M10	prefill	16,384	1129.638	0.567	0.050	1
Mac Studio M4 Max	M12	decode	0	48.365	0.091	0.188	1
Mac Studio M4 Max	M12	decode	4,096	47.519	0.047	0.099	1
Mac Studio M4 Max	M12	decode	16,384	45.097	0.041	0.090	1
Mac Studio M4 Max	M12	prefill	0	880.266	0.530	0.060	1
Mac Studio M4 Max	M12	prefill	4,096	834.386	0.599	0.072	1
Mac Studio M4 Max	M12	prefill	16,384	722.102	0.565	0.078	1
Mac Studio M4 Max	M14	decode	0	15.697	0.063	0.403	1
Mac Studio M4 Max	M14	decode	4,096	15.505	0.041	0.266	1
Mac Studio M4 Max	M14	decode	16,384	14.931	0.022	0.150	1
Mac Studio M4 Max	M14	prefill	0	255.725	0.175	0.068	1
Mac Studio M4 Max	M14	prefill	4,096	244.576	0.168	0.069	1
Mac Studio M4 Max	M14	prefill	16,384	215.717	0.047	0.022	1

Probe-exclusion sensitivity. For each host with calibration probes, B2 is refit after adding those files back. The original-train column evaluates that refit only on non-probe training rows; test rows remain held out.

Host	Probe cfg.	Aug. train n	Aug. MAPE	Original n	Original MAPE	Test n	Test MAPE
MacBook Pro M4 Max	3	54	10.46	45	11.06	12	13.11
Mac Studio M4 Max	3	54	12.65	45	13.39	15	14.37

2 Aggregate prediction results

Results are separated by host because fitted efficiency factors are host-specific; a pooled row would weight the unequal host cohorts and is not a cross-system score.

Host	Predictor	Split	n	MAPE (%)	Median (%)	P90 / max (%)
MacBook Pro M4 Max	B0 unfitted (eta=1, total)	train	45	33.57	19.37	76.47 / 123.27
MacBook Pro M4 Max	B0 unfitted (eta=1, total)	test	12	46.89	49.97	83.45 / 84.22
MacBook Pro M4 Max	B1 fitted, total params	train	45	15.80	3.20	70.29 / 76.77
MacBook Pro M4 Max	B1 fitted, total params	test	12	49.36	54.66	81.90 / 82.74
MacBook Pro M4 Max	B2 fitted, active params (ours)	train	45	10.39	2.12	26.32 / 78.49
MacBook Pro M4 Max	B2 fitted, active params (ours)	test	12	13.11	8.98	36.65 / 49.92
Mac Studio M4 Max	B0 unfitted (eta=1, total)	train	45	32.95	19.40	75.52 / 125.41
Mac Studio M4 Max	B0 unfitted (eta=1, total)	test	15	57.98	81.60	92.02 / 92.67
Mac Studio M4 Max	B1 fitted, total params	train	45	17.76	7.49	68.32 / 75.84
Mac Studio M4 Max	B1 fitted, total params	test	15	55.25	72.08	80.95 / 81.72
Mac Studio M4 Max	B2 fitted, active params (ours)	train	45	12.40	7.49	29.09 / 78.32
Mac Studio M4 Max	B2 fitted, active params (ours)	test	15	14.37	11.30	33.89 / 52.62
RTX 5080	B0 unfitted (eta=1, total)	train	36	20.27	14.47	44.15 / 71.19
RTX 5080	B0 unfitted (eta=1, total)	test	6	41.65	40.39	81.22 / 81.95
RTX 5080	B1 fitted, total params	train	36	9.85	8.88	20.95 / 31.73
RTX 5080	B1 fitted, total params	test	6	51.85	50.81	84.50 / 85.11
RTX 5080	B2 fitted, active params (ours)	train	36	10.12	9.34	20.95 / 31.73
RTX 5080	B2 fitted, active params (ours)	test	6	36.15	25.79	62.41 / 68.96
MacBook Pro M4 Max	P1 eta_p per host	train	15	6.81	5.36	16.58 / 33.40
MacBook Pro M4 Max	P1 eta_p per host	test	4	22.68	19.16	44.95 / 51.25
MacBook Pro M4 Max	P2 eta_p per host x quant	train	15	4.13	0.00	15.06 / 25.99
MacBook Pro M4 Max	P2 eta_p per host x quant	test	4	18.68	14.90	36.90 / 42.85
Mac Studio M4 Max	P1 eta_p per host	train	15	7.00	2.17	18.89 / 36.84
Mac Studio M4 Max	P1 eta_p per host	test	5	28.04	30.15	46.54 / 47.19
Mac Studio M4 Max	P2 eta_p per host x quant	train	15	4.27	0.00	16.14 / 26.29
Mac Studio M4 Max	P2 eta_p per host x quant	test	5	22.23	20.11	40.03 / 42.81
RTX 5080	P1 eta_p per host	train	12	14.46	9.15	31.38 / 52.16
RTX 5080	P1 eta_p per host	test	2	124.51	124.51	207.24 / 227.93
RTX 5080	P2 eta_p per host x quant	train	12	5.86	0.00	21.31 / 25.36
RTX 5080	P2 eta_p per host x quant	test	2	108.18	108.18	184.89 / 204.07

Decode summaries use all three context depths. Prefill headline rows use only depth 0, because the stated prefill model has no existing-prefix term; deeper rows are scope diagnostics below.

2.1 Leave-one-host-out transfer

B2 fits median per-quantization coefficients on all non-target training hosts and substitutes the untouched target host’s bandwidth. The target set contains all eligible rows on that host; because most files also occur on at least one source host, this isolates host/runtime transfer rather than a simultaneous model-and-hardware holdout. A target format absent from all source training rows uses the source-wide median fallback. The rejected two-term rows are the absolute-time output-projection extension.

Variant	Target host	Source n	Target n	MAPE	Median	Max
B2: all target rows	MacBook Pro M4 Max	81	57	14.47	8.02	73.61
B2: all target rows	Mac Studio M4 Max	81	60	15.42	10.78	84.30
B2: all target rows	RTX 5080	90	42	20.80	18.18	68.13
B2: target test rows	MacBook Pro M4 Max	81	12	11.59	6.00	45.82
B2: target test rows	Mac Studio M4 Max	81	15	16.76	14.36	57.74
B2: target test rows	RTX 5080	90	6	35.97	25.66	68.13
rejected two-term	MacBook Pro M4 Max	81	57	17.88	12.10	75.66
rejected two-term	Mac Studio M4 Max	81	60	16.93	9.50	66.06
rejected two-term	RTX 5080	90	42	149.33	132.38	501.57

For context, target-fitted B2 gives all-row MAPE of 10.96% on MacBook Pro M4 Max; 12.89% on Mac Studio M4 Max; 13.84% on RTX 5080. Unlike the untouched-target transfer rows, these comparators mix training rows used to fit the target coefficients with held-out rows.

Host	Split	Architecture	n	B0	B1	B2
MacBook Pro M4 Max	train	dense	36	23.21	7.14	5.66
MacBook Pro M4 Max	train	MoE	9	74.98	50.45	29.29
MacBook Pro M4 Max	test	dense	6	11.18	17.75	3.28
MacBook Pro M4 Max	test	MoE	6	82.61	80.97	22.95
Mac Studio M4 Max	train	dense	36	22.74	9.91	7.89
Mac Studio M4 Max	train	MoE	9	73.79	49.15	30.45
Mac Studio M4 Max	test	dense	6	17.21	22.40	6.31
Mac Studio M4 Max	test	MoE	9	85.16	77.15	19.74
RTX 5080	train	dense	33	15.73	10.57	10.57
RTX 5080	train	MoE	3	70.28	1.95	5.18
RTX 5080	test	dense	3	3.54	20.40	20.40
RTX 5080	test	MoE	3	79.77	83.31	51.90

2.2 Per-layer KV correction

With each variant refit on training rows, replacing a uniform-global KV denominator by the frozen per-layer metadata produces:

Host	Split	n	Uniform KV MAPE (%)	Per-layer KV MAPE (%)
MacBook Pro M4 Max	train	45	12.68	10.39
MacBook Pro M4 Max	test	12	21.23	13.11
Mac Studio M4 Max	train	45	14.68	12.40
Mac Studio M4 Max	test	15	20.96	14.37
RTX 5080	train	36	12.37	10.12
RTX 5080	test	6	37.87	36.15

2.3 Decode context behavior

All 53 scored host–file configurations slow monotonically from 0 through 16,384 tokens. The normalized rate is each model’s throughput divided by its own depth-0 throughput.

Host	Depth	Train n	Test n	Train B2 MAPE	Test B2 MAPE	Median normalized rate
MacBook Pro M4 Max	0	15	4	11.30	17.86	1.000
MacBook Pro M4 Max	4,096	15	4	7.59	12.83	0.945
MacBook Pro M4 Max	16,384	15	4	12.27	8.63	0.837
Mac Studio M4 Max	0	15	5	13.60	18.13	1.000
Mac Studio M4 Max	4,096	15	5	8.07	15.28	0.976
Mac Studio M4 Max	16,384	15	5	15.54	9.69	0.915
RTX 5080	0	12	2	10.19	44.52	1.000
RTX 5080	4,096	12	2	4.49	38.28	0.985
RTX 5080	16,384	12	2	15.67	25.64	0.922

2.4 Prefill scope diagnostic

Depth 0 is the primary prefill fit and score. The same fitted coefficients are applied unchanged at larger existing-prefix depths. These errors are prediction errors, distinct from within-run prefill variability; results remain separated by host.

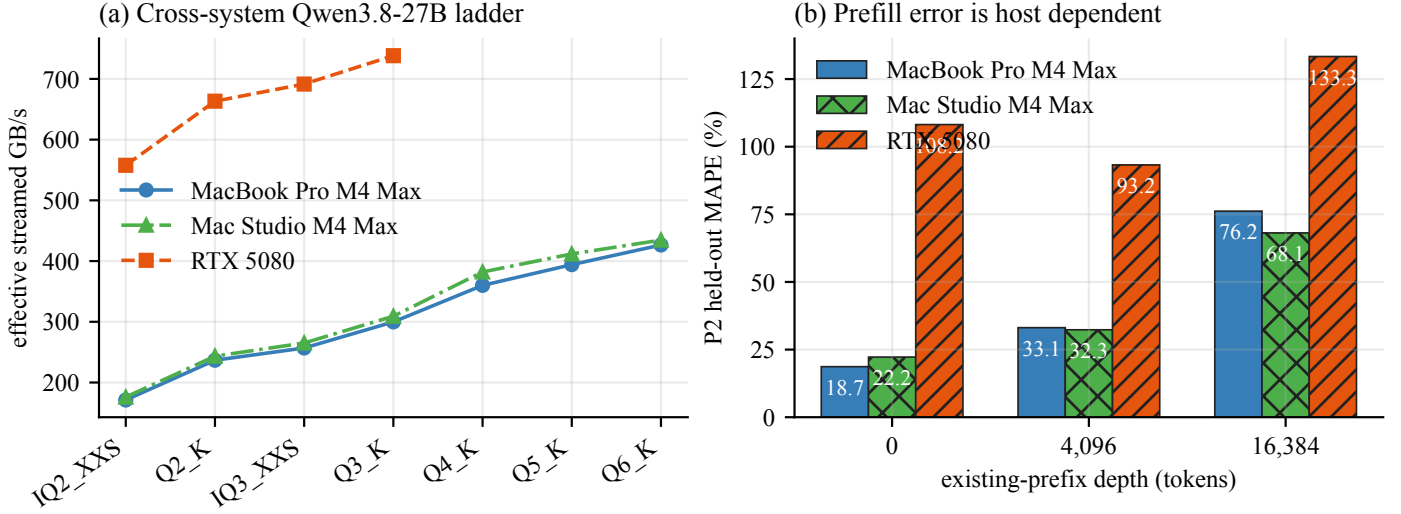


Figure 1: Host-separated diagnostics. Left: effective streamed bandwidth for the scored Qwen3.8-27B quantizations at zero prefix; host coverage differs, and IQ2 has 64 layers/26.90B parameters versus 65 layers/27.32B for the other shown files, so the set is a family comparison rather than a strictly identical-network quantization ladder. Right: P2 held-out error when depth-0 coefficients are applied at each existing-prefix depth. Host results are not pooled.

Host	Split	Depth	n	P1 MAPE	P2 MAPE	P2 median	P2 max
MacBook Pro M4 Max	train	0	15	6.81	4.13	0.00	25.99
MacBook Pro M4 Max	train	4,096	15	22.35	20.83	17.58	49.82
MacBook Pro M4 Max	train	16,384	15	66.94	64.13	51.02	150.97
MacBook Pro M4 Max	test	0	4	22.68	18.68	14.90	42.85
MacBook Pro M4 Max	test	4,096	4	40.98	33.11	29.60	55.36
MacBook Pro M4 Max	test	16,384	4	86.59	76.18	73.78	132.42
Mac Studio M4 Max	train	0	15	7.00	4.27	0.00	26.29
Mac Studio M4 Max	train	4,096	15	15.96	13.66	8.81	40.52
Mac Studio M4 Max	train	16,384	15	47.15	42.84	18.25	115.02
Mac Studio M4 Max	test	0	5	28.04	22.23	20.11	42.81
Mac Studio M4 Max	test	4,096	5	38.87	32.32	32.35	57.15
Mac Studio M4 Max	test	16,384	5	78.65	68.11	63.93	126.96
RTX 5080	train	0	12	14.46	5.86	0.00	25.36
RTX 5080	train	4,096	12	19.26	12.35	8.23	32.74
RTX 5080	train	16,384	12	36.13	28.56	18.19	72.26
RTX 5080	test	0	2	124.51	108.18	108.18	204.07
RTX 5080	test	4,096	2	108.39	93.23	93.23	167.57
RTX 5080	test	16,384	2	151.64	133.34	133.34	227.36

3 Selected model metadata

Tables 2 and 3 jointly reproduce every frozen metadata field used for the 23 unique selected files (22 scored and 1 probe-only). Scored host coverage is explicit because the measurement cohorts are not identical.

Table 2: Selected model identity, source, and file size. Sizes are exact bytes; host codes are MB (MacBook Pro M4 Max), MS (Mac Studio M4 Max), RTX (RTX 5080), and “probe only” denotes a file excluded from every prediction fit and score.

ID	Split	Scored host(s)	GGUF filename	Repository	Architecture	Quant.	Bytes
M01	train	MB, MS, RTX	gemma-4-12b-it-qat-q4_0.gguf	google/gemma-4-12b-it-qat-q4_0-gguf	gemma4	Q4_0	6,975,879,296
M02	train	MB, MS	gemma-4-26B-A4B-it-UD-Q4_K_M.gguf	unsloth/gemma-4-26B-A4B-it-GGUF	gemma4	Q4_K_M	16,947,541,728
M03	test	MS	gpt-oss-120b-MXFP4.gguf	ggml-org/gpt-oss-120b-GGUF	gpt-oss	MXFP4	63,387,346,208
M04	train	MB, MS, RTX	gpt-oss-20b-MXFP4.gguf	ggml-org/gpt-oss-20b-GGUF	gpt-oss	MXFP4	12,109,566,624
M05	test	MB, MS, RTX	inclusionAI_Ling-mini-2.0-Q4_K_M.gguf	bartowski/inclusionAI_Ling-mini-2.0-GGUF	bailingmoe2	Q4_K_M	9,941,894,336
M06	test	MB, MS	Muse-Glimmer-30B-UD-Q4_K_XL.gguf	unsloth/Muse-Glimmer-30B-GGUF	muse-glimmer	Q4_K	15,878,222,368
M07	test	MB, MS	Nemotron-3-Nano-30B-A3B-Q4_K_M.gguf	unsloth/Nemotron-3-Nano-30B-A3B-GGUF	nemotron_h_moe	Q4_K_M	24,574,373,664
M08	test	MB, MS, RTX	NVIDIA-Nemotron-3-Nano-4B-Q4_K_M.gguf	lmstudio-community/NVIDIA-Nemotron-3-Nano-4B-GGUF	nemotron_h	Q4_K_M	2,837,072,896
M09	train	MB, MS, RTX	Qwen3.5-4B-Q4_K_M.gguf	unsloth/Qwen3.5-4B-GGUF	qwen35	Q4_K_M	2,740,937,888
M10	train	RTX	Qwen3.5-4B-Q8_0.gguf	unsloth/Qwen3.5-4B-GGUF	qwen35	Q8_0	4,482,403,488
M11	train	MB, MS, RTX	Qwen3.5-9B-Q4_K_M.gguf	unsloth/Qwen3.5-9B-GGUF	qwen35	Q4_K_M	5,680,522,464
M12	train	RTX	Qwen3.5-9B-Q8_0.gguf	unsloth/Qwen3.5-9B-GGUF	qwen35	Q8_0	9,527,502,048
M13	train	MB, MS	Qwen3.6-35B-A3B-UD-Q4_K_M.gguf	unsloth/Qwen3.6-35B-A3B-GGUF	qwen35moe	Q4_K_M	22,134,528,992
M14	probe only	–	Qwen3.8-27B-Q8_0.gguf	unsloth/Qwen3.8-27B-GGUF	qwen35	Q8_0	29,047,086,048
M15	train	MB, MS, RTX	Qwen3.8-27B-UD-IQ2_XXS.gguf	unsloth/Qwen3.8-27B-GGUF	qwen35	IQ2_XXS	7,266,070,528
M16	train	MB, MS, RTX	Qwen3.8-27B-UD-IQ3_XXS.gguf	unsloth/Qwen3.8-27B-GGUF	qwen35	IQ3_XXS	10,934,860,704
M17	train	MB, MS, RTX	Qwen3.8-27B-UD-Q2_K_XL.gguf	unsloth/Qwen3.8-27B-GGUF	qwen35	Q2_K	9,828,981,664
M18	train	MB, MS, RTX	Qwen3.8-27B-UD-Q3_K_XL.gguf	unsloth/Qwen3.8-27B-GGUF	qwen35	Q3_K	13,146,393,504
M19	train	MB, MS	Qwen3.8-27B-UD-Q4_K_XL.gguf	unsloth/Qwen3.8-27B-GGUF	qwen35	Q4_K	17,559,178,144
M20	train	MB, MS	Qwen3.8-27B-UD-Q5_K_XL.gguf	unsloth/Qwen3.8-27B-GGUF	qwen35	Q5_K	20,876,938,144
M21	train	MB, MS	Qwen3.8-27B-UD-Q6_K_XL.gguf	unsloth/Qwen3.8-27B-GGUF	qwen35	Q6_K	25,299,061,664
M22	train	MB, MS, RTX	SmolLM3-Q4_K_M.gguf	ggml-org/SmolLM3-3B-GGUF	smollm3	Q4_K_M	1,915,305,312
M23	train	MB, MS, RTX	SmolLM3-Q8_0.gguf	ggml-org/SmolLM3-3B-GGUF	smollm3	Q8_0	3,275,574,624

Table 3: Selected model geometry. KV columns are exact bytes at the measured depths; k/E is routed/total experts.

ID	Total P	Active P	Active %	Layers	Width	Vocab.	KV heads	Head dim.	Used/all	KV@0	KV@4096	KV@16384
M01	11,907,350,576	11,907,350,576	100.0	48	3,840	262,144	8	512	—	0	738,197,504	939,524,096
M02	25,233,142,046	3,822,527,246	15.1	30	2,816	262,144	8	512	8/128	0	503,316,480	754,974,720
M03	116,829,156,672	5,711,982,912	4.9	36	2,880	201,088	8	64	4/128	0	301,989,888	1,207,959,552
M04	20,914,757,184	4,187,440,704	20.0	24	2,880	201,088	8	64	4/32	0	201,326,592	805,306,368
M05	16,255,643,392	1,432,973,056	8.8	20	2,048	157,184	4	128	8/256	0	167,772,160	671,088,640
M06	27,854,794,240	27,854,794,240	100.0	52	6,656	202,048	2	128	—	0	218,103,808	872,415,232
M07	31,577,940,288	3,580,076,352	11.3	52	2,688	131,072	2	128	6/128	397,934,592	423,100,416	498,597,888
M08	3,973,556,832	3,973,556,832	100.0	42	3,136	131,072	8	128	—	616,366,080	683,474,944	884,801,536
M09	4,205,751,296	4,205,751,296	100.0	32	2,560	248,320	4	256	—	0	536,870,912	2,147,483,648
M10	4,205,751,296	4,205,751,296	100.0	32	2,560	248,320	4	256	—	0	536,870,912	2,147,483,648
M11	8,953,803,264	8,953,803,264	100.0	32	4,096	248,320	4	256	—	0	536,870,912	2,147,483,648
M12	8,953,803,264	8,953,803,264	100.0	32	4,096	248,320	4	256	—	0	536,870,912	2,147,483,648
M13	34,660,610,688	3,454,988,928	10.0	40	2,048	248,320	2	256	8/256	0	335,544,320	1,342,177,280
M14	27,320,697,856	27,320,697,856	100.0	65	5,120	248,320	4	256	—	0	1,090,519,040	4,362,076,160
M15	26,895,998,464	26,895,998,464	100.0	64	5,120	248,320	4	256	—	0	1,073,741,824	4,294,967,296
M16	27,320,697,856	27,320,697,856	100.0	65	5,120	248,320	4	256	—	0	1,090,519,040	4,362,076,160
M17	27,320,697,856	27,320,697,856	100.0	65	5,120	248,320	4	256	—	0	1,090,519,040	4,362,076,160
M18	27,320,697,856	27,320,697,856	100.0	65	5,120	248,320	4	256	—	0	1,090,519,040	4,362,076,160
M19	27,320,697,856	27,320,697,856	100.0	65	5,120	248,320	4	256	—	0	1,090,519,040	4,362,076,160
M20	27,320,697,856	27,320,697,856	100.0	65	5,120	248,320	4	256	—	0	1,090,519,040	4,362,076,160
M21	27,320,697,856	27,320,697,856	100.0	65	5,120	248,320	4	256	—	0	1,090,519,040	4,362,076,160
M22	3,075,098,624	3,075,098,624	100.0	36	2,048	128,256	4	128	—	0	301,989,888	1,207,959,552
M23	3,075,098,624	3,075,098,624	100.0	36	2,048	128,256	4	128	—	0	301,989,888	1,207,959,552

4 All scored decode observations

These are the 159 rows exported by results/predictions.csv. Throughput is tokens/s; APE is absolute percentage error. Rows are ordered by host, split, model ID, and depth.

Table 4: Protocol-complete decode measurements and B2 predictions. The retry gate uses the worst decode CV in each host–file invocation. Pre-load is the macOS one-minute POSIX load average or, on Windows, busy-logical-CPU equivalents; it is meaningful within a host but not comparable across hosts.

Host	ID	Split	Depth	Measured	B2	APE %	Row CV %	Load	Attempts
MacBook Pro M4 Max	M01	train	0	54.28	54.28	0.00	0.42	15.79	1
MacBook Pro M4 Max	M01	train	4,096	49.91	49.08	1.65	0.55	15.79	1
MacBook Pro M4 Max	M01	train	16,384	45.34	47.84	5.50	1.54	15.79	1
MacBook Pro M4 Max	M02	train	0	89.15	124.01	39.11	0.15	16.58	1
MacBook Pro M4 Max	M02	train	4,096	82.91	103.69	25.07	0.42	16.58	1
MacBook Pro M4 Max	M02	train	16,384	75.37	95.83	27.15	0.44	16.58	1
MacBook Pro M4 Max	M04	train	0	124.28	128.17	3.13	0.29	11.83	1
MacBook Pro M4 Max	M04	train	4,096	118.34	118.34	0.00	0.19	11.83	1
MacBook Pro M4 Max	M04	train	16,384	102.83	96.21	6.43	0.67	11.83	1
MacBook Pro M4 Max	M09	train	0	101.64	116.16	14.29	0.16	8.19	1
MacBook Pro M4 Max	M09	train	4,096	97.13	97.13	0.00	0.28	8.19	1
MacBook Pro M4 Max	M09	train	16,384	85.05	65.13	23.42	1.25	8.19	1
MacBook Pro M4 Max	M11	train	0	66.64	56.05	15.90	1.00	8.30	1
MacBook Pro M4 Max	M11	train	4,096	62.95	51.21	18.65	1.81	8.30	1
MacBook Pro M4 Max	M11	train	16,384	56.62	40.67	28.17	2.81	8.30	1
MacBook Pro M4 Max	M13	train	0	80.85	144.30	78.49	0.29	15.62	1
MacBook Pro M4 Max	M13	train	4,096	78.38	125.26	59.81	0.24	15.62	1
MacBook Pro M4 Max	M13	train	16,384	72.10	89.72	24.44	1.12	15.62	1
MacBook Pro M4 Max	M15	train	0	23.56	23.56	0.00	1.20	16.52	1
MacBook Pro M4 Max	M15	train	4,096	20.41	20.53	0.59	2.70	16.52	1
MacBook Pro M4 Max	M15	train	16,384	19.71	14.81	24.88	1.72	16.52	1
MacBook Pro M4 Max	M16	train	0	23.49	24.42	3.94	1.09	14.84	1
MacBook Pro M4 Max	M16	train	4,096	22.20	22.20	0.00	1.59	14.84	1
MacBook Pro M4 Max	M16	train	16,384	19.63	17.45	11.07	1.65	14.84	1
MacBook Pro M4 Max	M17	train	0	24.08	25.71	6.75	1.90	11.04	1
MacBook Pro M4 Max	M17	train	4,096	23.14	23.14	0.00	2.39	11.04	1
MacBook Pro M4 Max	M17	train	16,384	21.30	17.81	16.41	1.65	11.04	1
MacBook Pro M4 Max	M18	train	0	22.80	23.29	2.12	2.60	11.42	3
MacBook Pro M4 Max	M18	train	4,096	21.89	21.50	1.75	0.91	11.42	3
MacBook Pro M4 Max	M18	train	16,384	17.49	17.49	0.00	2.94	11.42	3
MacBook Pro M4 Max	M19	train	0	20.50	20.92	2.05	2.11	13.06	3

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Table 4: Protocol-complete decode measurements and B2 predictions. The retry gate uses the worst decode CV in each host–file invocation. Pre-load is the macOS one-minute POSIX load average or, on Windows, busy-logical-CPU equivalents; it is meaningful within a host but not comparable across hosts. (continued)

Host	ID	Split	Depth	Measured	B2	APE %	Row CV %	Load	Attempts
MacBook Pro M4 Max	M19	train	4,096	19.70	19.70	0.00	1.97	13.06	3
MacBook Pro M4 Max	M19	train	16,384	16.77	16.76	0.10	1.47	13.06	3
MacBook Pro M4 Max	M20	train	0	18.88	19.13	1.31	1.57	17.66	1
MacBook Pro M4 Max	M20	train	4,096	18.33	18.18	0.83	2.21	17.66	1
MacBook Pro M4 Max	M20	train	16,384	15.82	15.82	0.00	2.67	17.66	1
MacBook Pro M4 Max	M21	train	0	16.86	16.86	0.00	0.37	17.18	1
MacBook Pro M4 Max	M21	train	4,096	15.92	16.16	1.50	1.42	17.18	1
MacBook Pro M4 Max	M21	train	16,384	14.97	14.38	3.96	0.92	17.18	1
MacBook Pro M4 Max	M22	train	0	169.50	166.23	1.93	0.46	13.91	1
MacBook Pro M4 Max	M22	train	4,096	149.47	143.59	3.93	0.31	13.91	1
MacBook Pro M4 Max	M22	train	16,384	115.22	101.94	11.53	0.40	13.91	1
MacBook Pro M4 Max	M23	train	0	118.87	119.53	0.55	0.40	13.92	1
MacBook Pro M4 Max	M23	train	4,096	109.44	109.44	0.00	0.41	13.92	1
MacBook Pro M4 Max	M23	train	16,384	88.22	87.32	1.01	1.00	13.92	1
MacBook Pro M4 Max	M05	test	0	242.33	363.29	49.92	1.24	11.64	1
MacBook Pro M4 Max	M05	test	4,096	219.67	304.92	38.81	0.84	11.64	1
MacBook Pro M4 Max	M05	test	16,384	175.60	205.75	17.17	0.37	11.64	1
MacBook Pro M4 Max	M06	test	0	26.23	23.14	11.81	1.92	8.93	3
MacBook Pro M4 Max	M06	test	4,096	23.30	22.82	2.05	2.13	8.93	3
MacBook Pro M4 Max	M06	test	16,384	22.55	21.93	2.74	3.03	8.93	3
MacBook Pro M4 Max	M07	test	0	92.12	100.00	8.55	0.30	10.01	1
MacBook Pro M4 Max	M07	test	4,096	90.68	99.21	9.41	0.49	10.01	1
MacBook Pro M4 Max	M07	test	16,384	85.16	96.93	13.83	0.68	10.01	1
MacBook Pro M4 Max	M08	test	0	93.29	92.19	1.18	0.30	8.78	1
MacBook Pro M4 Max	M08	test	4,096	91.41	90.44	1.07	0.10	8.78	1
MacBook Pro M4 Max	M08	test	16,384	84.86	85.55	0.80	0.42	8.78	1
Mac Studio M4 Max	M01	train	0	54.46	54.46	0.00	0.41	10.38	1
Mac Studio M4 Max	M01	train	4,096	50.62	49.25	2.71	2.06	10.38	1
Mac Studio M4 Max	M01	train	16,384	47.37	47.99	1.31	0.16	10.38	1
Mac Studio M4 Max	M02	train	0	88.10	124.32	41.11	0.22	4.69	1
Mac Studio M4 Max	M02	train	4,096	79.77	103.94	30.31	2.30	4.69	1
Mac Studio M4 Max	M02	train	16,384	75.49	96.07	27.26	0.24	4.69	1
Mac Studio M4 Max	M04	train	0	123.10	126.93	3.11	0.36	3.28	1
Mac Studio M4 Max	M04	train	4,096	117.20	117.20	0.00	0.81	3.28	1
Mac Studio M4 Max	M04	train	16,384	105.36	95.28	9.57	0.31	3.28	1
Mac Studio M4 Max	M09	train	0	101.41	116.45	14.83	0.42	4.06	1

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Table 4: Protocol-complete decode measurements and B2 predictions. The retry gate uses the worst decode CV in each host–file invocation. Pre-load is the macOS one-minute POSIX load average or, on Windows, busy-logical-CPU equivalents; it is meaningful within a host but not comparable across hosts. (continued)

Host	ID	Split	Depth	Measured	B2	APE %	Row CV %	Load	Attempts
Mac Studio M4 Max	M09	train	4,096	97.37	97.37	0.00	1.90	4.06	1
Mac Studio M4 Max	M09	train	16,384	88.73	65.29	26.41	0.23	4.06	1
Mac Studio M4 Max	M11	train	0	68.03	56.19	17.41	0.12	3.12	1
Mac Studio M4 Max	M11	train	4,096	66.41	51.34	22.69	0.14	3.12	1
Mac Studio M4 Max	M11	train	16,384	61.83	40.77	34.06	0.18	3.12	1
Mac Studio M4 Max	M13	train	0	81.12	144.66	78.32	0.68	3.80	1
Mac Studio M4 Max	M13	train	4,096	77.79	125.56	61.42	1.17	3.80	1
Mac Studio M4 Max	M13	train	16,384	73.18	89.94	22.91	0.87	3.80	1
Mac Studio M4 Max	M15	train	0	24.22	27.31	12.74	0.37	2.98	1
Mac Studio M4 Max	M15	train	4,096	23.79	23.79	0.00	0.95	2.98	1
Mac Studio M4 Max	M15	train	16,384	22.40	17.16	23.36	0.30	2.98	1
Mac Studio M4 Max	M16	train	0	24.24	26.19	8.02	0.31	2.14	1
Mac Studio M4 Max	M16	train	4,096	23.81	23.81	0.00	0.69	2.14	1
Mac Studio M4 Max	M16	train	16,384	22.34	18.72	16.18	0.55	2.14	1
Mac Studio M4 Max	M17	train	0	24.76	26.91	8.68	0.45	2.92	1
Mac Studio M4 Max	M17	train	4,096	24.23	24.23	0.00	0.43	2.92	1
Mac Studio M4 Max	M17	train	16,384	22.83	18.64	18.35	0.71	2.92	1
Mac Studio M4 Max	M18	train	0	23.52	24.99	6.23	0.68	4.39	1
Mac Studio M4 Max	M18	train	4,096	23.07	23.07	0.00	0.47	4.39	1
Mac Studio M4 Max	M18	train	16,384	21.77	18.76	13.81	0.66	4.39	1
Mac Studio M4 Max	M19	train	0	21.76	22.86	5.07	1.00	6.44	1
Mac Studio M4 Max	M19	train	4,096	21.52	21.52	0.00	0.80	6.44	1
Mac Studio M4 Max	M19	train	16,384	20.41	18.31	10.29	1.01	6.44	1
Mac Studio M4 Max	M20	train	0	19.72	20.46	3.76	0.13	2.24	1
Mac Studio M4 Max	M20	train	4,096	19.45	19.45	0.00	0.33	2.24	1
Mac Studio M4 Max	M20	train	16,384	18.54	16.93	8.71	0.14	2.24	1
Mac Studio M4 Max	M21	train	0	17.18	17.67	2.87	0.38	2.99	1
Mac Studio M4 Max	M21	train	4,096	16.94	16.94	0.00	0.30	2.99	1
Mac Studio M4 Max	M21	train	16,384	16.29	15.07	7.49	0.17	2.99	1
Mac Studio M4 Max	M22	train	0	168.55	166.64	1.13	0.42	4.33	2
Mac Studio M4 Max	M22	train	4,096	149.85	143.95	3.94	1.20	4.33	2
Mac Studio M4 Max	M22	train	16,384	115.65	102.19	11.64	0.24	4.33	2
Mac Studio M4 Max	M23	train	0	119.69	120.58	0.75	0.25	5.97	1
Mac Studio M4 Max	M23	train	4,096	110.40	110.40	0.00	0.08	5.97	1
Mac Studio M4 Max	M23	train	16,384	89.60	88.10	1.68	2.06	5.97	1
Mac Studio M4 Max	M03	test	0	85.42	99.30	16.25	0.71	5.93	1

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Table 4: Protocol-complete decode measurements and B2 predictions. The retry gate uses the worst decode CV in each host–file invocation. Pre-load is the macOS one-minute POSIX load average or, on Windows, busy-logical-CPU equivalents; it is meaningful within a host but not comparable across hosts. (continued)

Host	ID	Split	Depth	Measured	B2	APE %	Row CV %	Load	Attempts
Mac Studio M4 Max	M03	test	4,096	81.13	90.48	11.54	1.09	5.93	1
Mac Studio M4 Max	M03	test	16,384	72.80	71.45	1.85	0.85	5.93	1
Mac Studio M4 Max	M05	test	0	238.62	364.19	52.62	1.31	3.14	1
Mac Studio M4 Max	M05	test	4,096	213.24	305.67	43.34	1.52	3.14	1
Mac Studio M4 Max	M05	test	16,384	172.28	206.25	19.72	0.74	3.14	1
Mac Studio M4 Max	M06	test	0	28.50	25.28	11.30	0.26	5.50	1
Mac Studio M4 Max	M06	test	4,096	27.82	24.94	10.35	0.58	5.50	1
Mac Studio M4 Max	M06	test	16,384	27.07	23.96	11.49	0.27	5.50	1
Mac Studio M4 Max	M07	test	0	91.77	100.24	9.24	0.66	2.65	1
Mac Studio M4 Max	M07	test	4,096	90.27	99.46	10.18	0.69	2.65	1
Mac Studio M4 Max	M07	test	16,384	86.07	97.17	12.90	0.47	2.65	1
Mac Studio M4 Max	M08	test	0	93.58	92.42	1.24	0.25	4.15	1
Mac Studio M4 Max	M08	test	4,096	91.56	90.66	0.99	0.82	4.15	1
Mac Studio M4 Max	M08	test	16,384	87.96	85.76	2.50	0.40	4.15	1
RTX 5080	M01	train	0	98.27	104.03	5.86	0.31	3.58	1
RTX 5080	M01	train	4,096	94.08	94.08	0.00	0.64	3.58	1
RTX 5080	M01	train	16,384	92.53	91.68	0.91	0.45	3.58	1
RTX 5080	M04	train	0	229.65	236.51	2.99	0.48	8.58	2
RTX 5080	M04	train	4,096	218.37	218.37	0.00	0.54	8.58	2
RTX 5080	M04	train	16,384	203.03	177.54	12.56	1.58	8.58	2
RTX 5080	M09	train	0	202.01	241.17	19.39	0.33	1.02	1
RTX 5080	M09	train	4,096	198.97	201.67	1.36	0.50	1.02	1
RTX 5080	M09	train	16,384	181.22	135.22	25.38	0.52	1.02	1
RTX 5080	M10	train	0	147.62	163.32	10.64	0.15	1.60	1
RTX 5080	M10	train	4,096	145.85	145.85	0.00	0.50	1.60	1
RTX 5080	M10	train	16,384	136.04	110.42	18.83	0.34	1.60	1
RTX 5080	M11	train	0	132.95	116.37	12.47	0.15	1.66	1
RTX 5080	M11	train	4,096	131.63	106.32	19.23	0.43	1.66	1
RTX 5080	M11	train	16,384	123.69	84.44	31.73	0.43	1.66	1
RTX 5080	M12	train	0	88.98	76.84	13.65	0.09	1.47	1
RTX 5080	M12	train	4,096	88.23	72.74	17.55	0.23	1.47	1
RTX 5080	M12	train	16,384	84.85	62.70	26.10	0.24	1.47	1
RTX 5080	M15	train	0	76.76	86.74	13.01	0.33	11.26	2
RTX 5080	M15	train	4,096	75.58	75.58	0.00	0.49	11.26	2
RTX 5080	M15	train	16,384	70.36	54.52	22.51	1.39	11.26	2
RTX 5080	M16	train	0	63.24	68.69	8.62	0.28	14.78	1

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Table 4: Protocol-complete decode measurements and B2 predictions. The retry gate uses the worst decode CV in each host–file invocation. Pre-load is the macOS one-minute POSIX load average or, on Windows, busy-logical-CPU equivalents; it is meaningful within a host but not comparable across hosts. (continued)

Host	ID	Split	Depth	Measured	B2	APE %	Row CV %	Load	Attempts
RTX 5080	M16	train	4,096	62.46	62.46	0.00	0.41	14.78	1
RTX 5080	M16	train	16,384	58.63	49.10	16.25	1.50	14.78	1
RTX 5080	M17	train	0	67.48	73.66	9.15	0.30	12.10	1
RTX 5080	M17	train	4,096	66.30	66.30	0.00	0.41	12.10	1
RTX 5080	M17	train	16,384	62.22	51.02	18.00	1.65	12.10	1
RTX 5080	M18	train	0	56.15	60.10	7.02	0.22	7.74	1
RTX 5080	M18	train	4,096	55.49	55.49	0.00	0.36	7.74	1
RTX 5080	M18	train	16,384	52.65	45.12	14.30	1.02	7.74	1
RTX 5080	M22	train	0	306.16	345.13	12.73	0.49	1.41	1
RTX 5080	M22	train	4,096	272.17	298.13	9.54	0.74	1.41	1
RTX 5080	M22	train	16,384	211.65	211.65	0.00	0.48	1.41	1
RTX 5080	M23	train	0	209.46	223.49	6.70	0.19	0.70	1
RTX 5080	M23	train	4,096	192.60	204.63	6.24	0.33	0.70	1
RTX 5080	M23	train	16,384	160.88	163.28	1.49	0.29	0.70	1
RTX 5080	M05	test	0	446.42	754.26	68.96	0.76	2.05	1
RTX 5080	M05	test	4,096	406.19	633.07	55.85	1.35	2.05	1
RTX 5080	M05	test	16,384	326.38	427.17	30.88	1.04	2.05	1
RTX 5080	M08	test	0	239.52	191.41	20.09	0.41	1.86	2
RTX 5080	M08	test	4,096	236.79	187.76	20.70	0.29	1.86	2
RTX 5080	M08	test	16,384	223.15	177.61	20.41	1.01	1.86	2

5 All scored prefill observations

These are the 159 rows exported by results/predictions_prefill.csv. P1 has one host coefficient; P2 has one host-by-quantization coefficient. Both are fit only at depth 0. The row CV column is reported rather than used as an inclusion filter.

Table 5: Protocol-complete prefill measurements and predictions. The retry decision was decode-gated, not prefill-gated; depth-0 rows are the primary evaluation and larger depths are scope diagnostics.

Host	ID	Split	Depth	Scope	Measured	P1	P1 APE	P2	P2 APE	Row CV %
MacBook Pro M4 Max	M01	train	0	primary	589.73	547.32	7.19	589.73	0.00	0.05
MacBook Pro M4 Max	M01	train	4,096	diagnostic	491.90	547.32	11.27	589.73	19.89	1.01
MacBook Pro M4 Max	M01	train	16,384	diagnostic	382.76	547.32	42.99	589.73	54.07	3.47
MacBook Pro M4 Max	M02	train	0	primary	1418.29	1704.93	20.21	1610.29	13.54	0.38
MacBook Pro M4 Max	M02	train	4,096	diagnostic	1145.69	1704.93	48.81	1610.29	40.55	2.81
MacBook Pro M4 Max	M02	train	16,384	diagnostic	741.02	1704.93	130.08	1610.29	117.31	2.63
MacBook Pro M4 Max	M04	train	0	primary	1589.84	1556.35	2.11	1589.84	0.00	0.27
MacBook Pro M4 Max	M04	train	4,096	diagnostic	1372.89	1556.35	13.36	1589.84	15.80	2.79
MacBook Pro M4 Max	M04	train	16,384	diagnostic	848.96	1556.35	83.33	1589.84	87.27	0.72
MacBook Pro M4 Max	M09	train	0	primary	1463.56	1549.58	5.88	1463.56	0.00	0.07
MacBook Pro M4 Max	M09	train	4,096	diagnostic	1310.45	1549.58	18.25	1463.56	11.68	0.40
MacBook Pro M4 Max	M09	train	16,384	diagnostic	941.78	1549.58	64.54	1463.56	55.40	0.89
MacBook Pro M4 Max	M11	train	0	primary	819.16	727.86	11.14	687.46	16.08	0.13
MacBook Pro M4 Max	M11	train	4,096	diagnostic	727.07	727.86	0.11	687.46	5.45	0.39
MacBook Pro M4 Max	M11	train	16,384	diagnostic	564.60	727.86	28.92	687.46	21.76	1.59
MacBook Pro M4 Max	M13	train	0	primary	1414.02	1886.30	33.40	1781.59	25.99	0.67
MacBook Pro M4 Max	M13	train	4,096	diagnostic	1189.17	1886.30	58.62	1781.59	49.82	1.38
MacBook Pro M4 Max	M13	train	16,384	diagnostic	880.48	1886.30	114.24	1781.59	102.34	1.39
MacBook Pro M4 Max	M15	train	0	primary	239.52	242.31	1.17	239.52	0.00	2.73
MacBook Pro M4 Max	M15	train	4,096	diagnostic	196.04	242.31	23.60	239.52	22.18	0.31
MacBook Pro M4 Max	M15	train	16,384	diagnostic	158.60	242.31	52.78	239.52	51.02	1.33
MacBook Pro M4 Max	M16	train	0	primary	238.54	238.54	0.00	238.54	0.00	2.42
MacBook Pro M4 Max	M16	train	4,096	diagnostic	203.40	238.54	17.28	238.54	17.28	1.81
MacBook Pro M4 Max	M16	train	16,384	diagnostic	172.91	238.54	37.96	238.54	37.96	1.74
MacBook Pro M4 Max	M17	train	0	primary	235.14	238.54	1.45	235.14	0.00	1.72
MacBook Pro M4 Max	M17	train	4,096	diagnostic	207.61	238.54	14.90	235.14	13.26	2.63
MacBook Pro M4 Max	M17	train	16,384	diagnostic	180.12	238.54	32.43	235.14	30.54	1.42
MacBook Pro M4 Max	M18	train	0	primary	241.51	238.54	1.23	241.51	0.00	2.59
MacBook Pro M4 Max	M18	train	4,096	diagnostic	202.71	238.54	17.67	241.51	19.14	0.93
MacBook Pro M4 Max	M18	train	16,384	diagnostic	164.64	238.54	44.89	241.51	46.69	2.76
MacBook Pro M4 Max	M19	train	0	primary	224.91	238.54	6.06	224.91	0.00	3.91
MacBook Pro M4 Max	M19	train	4,096	diagnostic	191.29	238.54	24.70	224.91	17.58	0.80

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Table 5: Protocol-complete prefill measurements and predictions. The retry decision was decode-gated, not prefill-gated; depth-0 rows are the primary evaluation and larger depths are scope diagnostics. (continued)

Host	ID	Split	Depth	Scope	Measured	P1	P1 APE	P2	P2 APE	Row CV %
MacBook Pro M4 Max	M19	train	16,384	diagnostic	162.37	238.54	46.91	224.91	38.52	2.61
MacBook Pro M4 Max	M20	train	0	primary	225.20	238.54	5.92	225.20	0.00	2.87
MacBook Pro M4 Max	M20	train	4,096	diagnostic	196.00	238.54	21.70	225.20	14.90	2.15
MacBook Pro M4 Max	M20	train	16,384	diagnostic	174.79	238.54	36.47	225.20	28.84	1.41
MacBook Pro M4 Max	M21	train	0	primary	239.03	238.54	0.20	239.03	0.00	2.54
MacBook Pro M4 Max	M21	train	4,096	diagnostic	208.07	238.54	14.64	239.03	14.88	2.22
MacBook Pro M4 Max	M21	train	16,384	diagnostic	181.38	238.54	31.52	239.03	31.78	1.35
MacBook Pro M4 Max	M22	train	0	primary	2135.94	2119.33	0.78	2001.68	6.29	0.04
MacBook Pro M4 Max	M22	train	4,096	diagnostic	1653.76	2119.33	28.15	2001.68	21.04	2.82
MacBook Pro M4 Max	M22	train	16,384	diagnostic	965.07	2119.33	119.60	2001.68	107.41	1.34
MacBook Pro M4 Max	M23	train	0	primary	2239.31	2119.33	5.36	2239.31	0.00	0.12
MacBook Pro M4 Max	M23	train	4,096	diagnostic	1735.34	2119.33	22.13	2239.31	29.04	0.12
MacBook Pro M4 Max	M23	train	16,384	diagnostic	892.27	2119.33	137.52	2239.31	150.97	1.34
MacBook Pro M4 Max	M05	test	0	primary	4208.20	4547.98	8.07	4295.53	2.08	0.42
MacBook Pro M4 Max	M05	test	4,096	diagnostic	3300.39	4547.98	37.80	4295.53	30.15	0.62
MacBook Pro M4 Max	M05	test	16,384	diagnostic	1848.16	4547.98	146.08	4295.53	132.42	1.94
MacBook Pro M4 Max	M06	test	0	primary	236.65	233.97	1.13	220.60	6.78	3.52
MacBook Pro M4 Max	M06	test	4,096	diagnostic	187.15	233.97	25.01	220.60	17.87	3.95
MacBook Pro M4 Max	M06	test	16,384	diagnostic	176.87	233.97	32.28	220.60	24.72	1.62
MacBook Pro M4 Max	M07	test	0	primary	1203.59	1820.39	51.25	1719.34	42.85	0.45
MacBook Pro M4 Max	M07	test	4,096	diagnostic	1106.68	1820.39	64.49	1719.34	55.36	3.64
MacBook Pro M4 Max	M07	test	16,384	diagnostic	895.52	1820.39	103.28	1719.34	91.99	1.19
MacBook Pro M4 Max	M08	test	0	primary	1259.18	1640.13	30.25	1549.08	23.02	4.84
MacBook Pro M4 Max	M08	test	4,096	diagnostic	1200.41	1640.13	36.63	1549.08	29.05	3.30
MacBook Pro M4 Max	M08	test	16,384	diagnostic	995.75	1640.13	64.71	1549.08	55.57	1.37
Mac Studio M4 Max	M01	train	0	primary	588.97	565.21	4.04	588.97	0.00	0.45
Mac Studio M4 Max	M01	train	4,096	diagnostic	521.77	565.21	8.32	588.97	12.88	0.20
Mac Studio M4 Max	M01	train	16,384	diagnostic	413.59	565.21	36.66	588.97	42.41	0.91
Mac Studio M4 Max	M02	train	0	primary	1408.11	1760.64	25.04	1624.90	15.40	0.75
Mac Studio M4 Max	M02	train	4,096	diagnostic	1181.84	1760.64	48.97	1624.90	37.49	0.93
Mac Studio M4 Max	M02	train	16,384	diagnostic	869.98	1760.64	102.38	1624.90	86.77	0.36
Mac Studio M4 Max	M04	train	0	primary	1576.97	1607.21	1.92	1576.97	0.00	0.79
Mac Studio M4 Max	M04	train	4,096	diagnostic	1389.95	1607.21	15.63	1576.97	13.45	0.27
Mac Studio M4 Max	M04	train	16,384	diagnostic	995.24	1607.21	61.49	1576.97	58.45	0.28
Mac Studio M4 Max	M09	train	0	primary	1476.84	1600.21	8.35	1476.84	0.00	0.12
Mac Studio M4 Max	M09	train	4,096	diagnostic	1357.24	1600.21	17.90	1476.84	8.81	0.20
Mac Studio M4 Max	M09	train	16,384	diagnostic	1083.61	1600.21	47.67	1476.84	36.29	0.16

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Table 5: Protocol-complete prefill measurements and predictions. The retry decision was decode-gated, not prefill-gated; depth-0 rows are the primary evaluation and larger depths are scope diagnostics. (continued)

Host	ID	Split	Depth	Scope	Measured	P1	P1 APE	P2	P2 APE	Row CV %
Mac Studio M4 Max	M11	train	0	primary	832.13	751.65	9.67	693.70	16.64	0.10
Mac Studio M4 Max	M11	train	4,096	diagnostic	792.01	751.65	5.10	693.70	12.41	0.12
Mac Studio M4 Max	M11	train	16,384	diagnostic	691.18	751.65	8.75	693.70	0.36	0.12
Mac Studio M4 Max	M13	train	0	primary	1423.47	1947.94	36.84	1797.76	26.29	0.75
Mac Studio M4 Max	M13	train	4,096	diagnostic	1279.33	1947.94	52.26	1797.76	40.52	0.47
Mac Studio M4 Max	M13	train	16,384	diagnostic	994.74	1947.94	95.82	1797.76	80.73	0.21
Mac Studio M4 Max	M15	train	0	primary	251.74	250.23	0.60	251.74	0.00	0.07
Mac Studio M4 Max	M15	train	4,096	diagnostic	240.96	250.23	3.84	251.74	4.47	0.03
Mac Studio M4 Max	M15	train	16,384	diagnostic	213.03	250.23	17.46	251.74	18.17	0.07
Mac Studio M4 Max	M16	train	0	primary	250.39	246.34	1.62	250.39	0.00	0.03
Mac Studio M4 Max	M16	train	4,096	diagnostic	239.79	246.34	2.73	250.39	4.42	0.04
Mac Studio M4 Max	M16	train	16,384	diagnostic	211.89	246.34	16.26	250.39	18.17	0.03
Mac Studio M4 Max	M17	train	0	primary	251.10	246.34	1.90	251.10	0.00	0.03
Mac Studio M4 Max	M17	train	4,096	diagnostic	240.25	246.34	2.53	251.10	4.52	0.06
Mac Studio M4 Max	M17	train	16,384	diagnostic	212.36	246.34	16.00	251.10	18.25	0.01
Mac Studio M4 Max	M18	train	0	primary	249.04	246.34	1.09	249.04	0.00	0.02
Mac Studio M4 Max	M18	train	4,096	diagnostic	238.52	246.34	3.28	249.04	4.41	0.04
Mac Studio M4 Max	M18	train	16,384	diagnostic	211.04	246.34	16.73	249.04	18.01	0.02
Mac Studio M4 Max	M19	train	0	primary	236.56	246.34	4.13	236.56	0.00	0.07
Mac Studio M4 Max	M19	train	4,096	diagnostic	227.01	246.34	8.51	236.56	4.21	0.06
Mac Studio M4 Max	M19	train	16,384	diagnostic	201.87	246.34	22.03	236.56	17.19	0.10
Mac Studio M4 Max	M20	train	0	primary	233.53	246.34	5.48	233.53	0.00	0.02
Mac Studio M4 Max	M20	train	4,096	diagnostic	224.14	246.34	9.90	233.53	4.19	0.05
Mac Studio M4 Max	M20	train	16,384	diagnostic	199.62	246.34	23.40	233.53	16.99	0.07
Mac Studio M4 Max	M21	train	0	primary	246.34	246.34	0.00	246.34	0.00	0.25
Mac Studio M4 Max	M21	train	4,096	diagnostic	236.32	246.34	4.24	246.34	4.24	0.09
Mac Studio M4 Max	M21	train	16,384	diagnostic	209.32	246.34	17.68	246.34	17.68	0.04
Mac Studio M4 Max	M22	train	0	primary	2142.88	2188.58	2.13	2019.85	5.74	0.26
Mac Studio M4 Max	M22	train	4,096	diagnostic	1678.58	2188.58	30.38	2019.85	20.33	0.26
Mac Studio M4 Max	M22	train	16,384	diagnostic	1019.72	2188.58	114.63	2019.85	98.08	0.10
Mac Studio M4 Max	M23	train	0	primary	2237.19	2188.58	2.17	2237.19	0.00	0.31
Mac Studio M4 Max	M23	train	4,096	diagnostic	1740.63	2188.58	25.73	2237.19	28.53	0.23
Mac Studio M4 Max	M23	train	16,384	diagnostic	1040.47	2188.58	110.35	2237.19	115.02	0.13
Mac Studio M4 Max	M03	test	0	primary	809.51	1178.24	45.55	1156.07	42.81	0.96
Mac Studio M4 Max	M03	test	4,096	diagnostic	735.66	1178.24	60.16	1156.07	57.15	0.61
Mac Studio M4 Max	M03	test	16,384	diagnostic	563.22	1178.24	109.20	1156.07	105.26	0.35
Mac Studio M4 Max	M05	test	0	primary	4177.10	4696.60	12.44	4334.51	3.77	0.74

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Table 5: Protocol-complete prefill measurements and predictions. The retry decision was decode-gated, not prefill-gated; depth-0 rows are the primary evaluation and larger depths are scope diagnostics. (continued)

Host	ID	Split	Depth	Scope	Measured	P1	P1 APE	P2	P2 APE	Row CV %
Mac Studio M4 Max	M05	test	4,096	diagnostic	3275.07	4696.60	43.40	4334.51	32.35	0.84
Mac Studio M4 Max	M05	test	16,384	diagnostic	1909.80	4696.60	145.92	4334.51	126.96	0.44
Mac Studio M4 Max	M06	test	0	primary	253.94	241.61	4.85	232.03	8.63	0.10
Mac Studio M4 Max	M06	test	4,096	diagnostic	241.32	241.61	0.12	232.03	3.85	0.10
Mac Studio M4 Max	M06	test	16,384	diagnostic	227.12	241.61	6.38	232.03	2.16	0.08
Mac Studio M4 Max	M07	test	0	primary	1277.14	1879.88	47.19	1734.94	35.85	0.19
Mac Studio M4 Max	M07	test	4,096	diagnostic	1213.08	1879.88	54.97	1734.94	43.02	0.23
Mac Studio M4 Max	M07	test	16,384	diagnostic	1058.33	1879.88	77.63	1734.94	63.93	0.18
Mac Studio M4 Max	M08	test	0	primary	1301.39	1693.72	30.15	1563.14	20.11	0.34
Mac Studio M4 Max	M08	test	4,096	diagnostic	1248.09	1693.72	35.71	1563.14	25.24	0.16
Mac Studio M4 Max	M08	test	16,384	diagnostic	1099.00	1693.72	54.11	1563.14	42.23	0.09
RTX 5080	M01	train	0	primary	5624.55	4619.99	17.86	5624.55	0.00	5.17
RTX 5080	M01	train	4,096	diagnostic	5030.69	4619.99	8.16	5624.55	11.80	3.92
RTX 5080	M01	train	16,384	diagnostic	4336.80	4619.99	6.53	5624.55	29.69	2.60
RTX 5080	M04	train	0	primary	8633.84	13137.34	52.16	8633.84	0.00	3.75
RTX 5080	M04	train	4,096	diagnostic	8160.96	13137.34	60.98	8633.84	5.79	4.00
RTX 5080	M04	train	16,384	diagnostic	7086.04	13137.34	85.40	8633.84	21.84	2.75
RTX 5080	M09	train	0	primary	9911.27	13080.14	31.97	12128.70	22.37	5.67
RTX 5080	M09	train	4,096	diagnostic	9373.61	13080.14	39.54	12128.70	29.39	4.03
RTX 5080	M09	train	16,384	diagnostic	8143.16	13080.14	60.63	12128.70	48.94	3.62
RTX 5080	M10	train	0	primary	10379.35	13080.14	26.02	13011.54	25.36	5.85
RTX 5080	M10	train	4,096	diagnostic	9802.11	13080.14	33.44	13011.54	32.74	4.92
RTX 5080	M10	train	16,384	diagnostic	8448.91	13080.14	54.81	13011.54	54.00	3.69
RTX 5080	M11	train	0	primary	6452.64	6143.96	4.78	5697.05	11.71	3.37
RTX 5080	M11	train	4,096	diagnostic	6219.18	6143.96	1.21	5697.05	8.40	2.66
RTX 5080	M11	train	16,384	diagnostic	5738.49	6143.96	7.07	5697.05	0.72	2.17
RTX 5080	M12	train	0	primary	6861.31	6143.96	10.45	6111.74	10.92	4.42
RTX 5080	M12	train	4,096	diagnostic	6647.95	6143.96	7.58	6111.74	8.07	2.90
RTX 5080	M12	train	16,384	diagnostic	6079.76	6143.96	1.06	6111.74	0.53	2.86
RTX 5080	M15	train	0	primary	1725.38	2045.35	18.54	1725.38	0.00	3.23
RTX 5080	M15	train	4,096	diagnostic	1682.57	2045.35	21.56	1725.38	2.54	1.97
RTX 5080	M15	train	16,384	diagnostic	1538.25	2045.35	32.97	1725.38	12.17	2.17
RTX 5080	M16	train	0	primary	2024.12	2013.56	0.52	2024.12	0.00	3.50
RTX 5080	M16	train	4,096	diagnostic	1954.57	2013.56	3.02	2024.12	3.56	2.46
RTX 5080	M16	train	16,384	diagnostic	1771.53	2013.56	13.66	2024.12	14.26	2.22
RTX 5080	M17	train	0	primary	2043.07	2013.56	1.44	2043.07	0.00	3.11
RTX 5080	M17	train	4,096	diagnostic	1962.09	2013.56	2.62	2043.07	4.13	3.14

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Table 5: Protocol-complete prefill measurements and predictions. The retry decision was decode-gated, not prefill-gated; depth-0 rows are the primary evaluation and larger depths are scope diagnostics. (continued)

Host	ID	Split	Depth	Scope	Measured	P1	P1 APE	P2	P2 APE	Row CV %
RTX 5080	M17	train	16,384	diagnostic	1783.83	2013.56	12.88	2043.07	14.53	2.23
RTX 5080	M18	train	0	primary	2042.02	2013.56	1.39	2042.02	0.00	17.04
RTX 5080	M18	train	4,096	diagnostic	2114.20	2013.56	4.76	2042.02	3.41	2.90
RTX 5080	M18	train	16,384	diagnostic	1907.04	2013.56	5.59	2042.02	7.08	2.39
RTX 5080	M22	train	0	primary	16588.19	17889.45	7.84	16588.19	0.00	7.30
RTX 5080	M22	train	4,096	diagnostic	14035.04	17889.45	27.46	16588.19	18.19	5.06
RTX 5080	M22	train	16,384	diagnostic	9948.68	17889.45	79.82	16588.19	66.74	3.46
RTX 5080	M23	train	0	primary	17795.63	17889.45	0.53	17795.63	0.00	8.37
RTX 5080	M23	train	4,096	diagnostic	14813.44	17889.45	20.76	17795.63	20.13	5.96
RTX 5080	M23	train	16,384	diagnostic	10330.65	17889.45	73.17	17795.63	72.26	3.83
RTX 5080	M05	test	0	primary	11706.91	38390.00	227.93	35597.53	204.07	39.19
RTX 5080	M05	test	4,096	diagnostic	13303.93	38390.00	188.56	35597.53	167.57	5.45
RTX 5080	M05	test	16,384	diagnostic	10873.99	38390.00	253.04	35597.53	227.36	4.90
RTX 5080	M08	test	0	primary	11433.58	13844.48	21.09	12837.44	12.28	4.86
RTX 5080	M08	test	4,096	diagnostic	10797.04	13844.48	28.22	12837.44	18.90	3.44
RTX 5080	M08	test	16,384	diagnostic	9215.27	13844.48	50.23	12837.44	39.31	3.36

6 Repeatability and calibration

A separate repeatability series exists only for the MacBook Pro M4 Max; variability on the other hosts is represented by the within-invocation repetitions in the row tables, not by equivalent across-run series. The six Mac runs give mean decode throughput 166.457 tokens/s and sample between-run CV 1.613%; prefill is 2133.091 tokens/s with CV 0.373%. The max-minus-min spans are 4.34% and 0.97% of their respective means. Decode has a descriptive linear trend of 0.74% per run. These are single-cell diagnostics, not general detection thresholds.

Run	Timestamp	Settle (s)	Prefill	Decode	Decode within-run SD
1	2026-08-30T01:06:08.096930-07:00	45	2137.572	164.323	6.520
2	2026-08-30T01:06:59.837356-07:00	45	2136.650	165.447	1.473
3	2026-08-30T01:07:51.671530-07:00	45	2136.742	163.200	10.629
4	2026-08-30T01:08:43.573411-07:00	45	2116.939	166.925	2.338
5	2026-08-30T01:09:35.318847-07:00	45	2135.238	168.415	0.805
6	2026-08-30T01:10:26.991582-07:00	45	2135.403	170.432	0.464

The append-only Mac measurement file also preserves a complete pre-protocol decode grid. It is excluded from every fit and score above. The comparison is MacBook-only and is not pooled with other hosts.

Pass	All <i>n</i>	Test <i>n</i>	Test MAPE	Median	P90	Max	All-row CV med.	All-row CV max
Archived pre-protocol	57	12	19.30	15.77	25.94	74.59	1.74	14.91
Final Mac protocol-complete	57	12	13.11	8.98	36.65	49.92	1.00	3.03

For comparison, the retained historical uncontrolled file contains two depth-0 observations of the same Qwen3.5-4B Q4_K_M cell:

Phase	Higher throughput	Lower throughput	Decrease (%)
prefill	1467.424	1090.676	25.67
decode	103.616	87.594	15.46

6.1 Host settings and calibration

Host	Platform	Accelerator	Threads	Reps	Settle (s)	Requested ngl
MacBook Pro M4 Max	Darwin-arm64	arm (Metal)	12	5	45	99
Mac Studio M4 Max	Darwin-arm64	arm (Metal)	12	5	45	99
RTX 5080	Windows-AMD64	NVIDIA GeForce RTX 5080	16	5	45	99

The requested `n_gpu_layers=99` setting asks the runtime to offload as many layers as it can; it is not direct telemetry proving that every tensor and KV allocation remained resident on the accelerator. No device-memory trace was recorded, so the appendix does not label these observations as confirmed fully resident.

Host	Calibration	Value	Detail
MacBook Pro M4 Max	CPU STREAM-style triad	48.558 GB/s	7 repetitions
MacBook Pro M4 Max	CPU FP32 matrix multiply	2.458 TFLOP/s	5 repetitions
MacBook Pro M4 Max	MPS device copy	380.052 GB/s	arm (Metal)
MacBook Pro M4 Max	MPS FP16 matrix multiply	13.950 TFLOP/s	arm (Metal)
MacBook Pro M4 Max	LLM reference ceiling	377.244 GB/s	Qwen3.5-9B-Q8_0.gguf
Mac Studio M4 Max	CPU STREAM-style triad	77.670 GB/s	7 repetitions
Mac Studio M4 Max	CPU FP32 matrix multiply	3.017 TFLOP/s	5 repetitions
Mac Studio M4 Max	MPS device copy	396.727 GB/s	arm (Metal)
Mac Studio M4 Max	MPS FP16 matrix multiply	15.001 TFLOP/s	arm (Metal)
Mac Studio M4 Max	LLM reference ceiling	425.530 GB/s	Qwen3.8-27B-Q8_0.gguf
RTX 5080	CPU STREAM-style triad	13.458 GB/s	7 repetitions
RTX 5080	CPU FP32 matrix multiply	2.314 TFLOP/s	5 repetitions
RTX 5080	CUDA device copy	801.061 GB/s	NVIDIA GeForce RTX 5080
RTX 5080	CUDA FP16 matrix multiply	118.829 TFLOP/s	NVIDIA GeForce RTX 5080

For the fitted B1/B2 and P1/P2 models, the measured bandwidth or FLOP/s scale cancels algebraically after fitting on this same host. It sets the scale of B0 and supports roofline interpretation; it does not by itself produce the reported fitted-model accuracy. The exported row-level bandwidth field uses the device-copy result on all hosts. The LLM references are retained as diagnostics and are not the B0 bandwidth value used in the current exports.

Host	Probe	File GB	Decode	SD	CV (%)	Effective GB/s
MacBook Pro M4 Max	Qwen3.8-27B-Q8_0.gguf	29.05	7.100	4.199	59.14	196.653
MacBook Pro M4 Max	Qwen3.5-9B-Q8_0.gguf	9.53	44.661	0.586	1.31	377.244
MacBook Pro M4 Max	Qwen3.5-4B-Q8_0.gguf	4.48	70.817	5.524	7.80	317.432
Mac Studio M4 Max	Qwen3.8-27B-Q8_0.gguf	29.05	15.364	0.080	0.52	425.530
Mac Studio M4 Max	Qwen3.5-9B-Q8_0.gguf	9.53	48.773	0.159	0.33	411.974
Mac Studio M4 Max	Qwen3.5-4B-Q8_0.gguf	4.48	79.528	0.146	0.18	356.475

7 Output-projection sensitivity

The additive output-projection term is reported as a sensitivity analysis. It is not selected: held-out improvement is not consistent across loss functions and hosts, and the term partly recharges weights already included in the streamed-byte term. Host rows and the unequally weighted all-host aggregate are shown separately. The one-term rows in this section are refit under the listed time-domain losses; they are not the median-ratio B2 headline and need not reproduce its MAPE.

Host	Loss	Model	Split	n	MAPE	Median	P90	Max
MacBook Pro M4 Max	absolute time	one term	train	45	12.38	4.97	34.07	91.71
MacBook Pro M4 Max	absolute time	one term	test	12	19.26	14.43	46.77	61.03
MacBook Pro M4 Max	absolute time	two terms	train	45	10.78	4.47	32.04	77.90
MacBook Pro M4 Max	absolute time	two terms	test	12	38.56	35.19	71.12	82.26
Mac Studio M4 Max	absolute time	one term	train	45	15.32	8.29	38.34	96.54
Mac Studio M4 Max	absolute time	one term	test	15	20.84	15.60	47.57	68.22
Mac Studio M4 Max	absolute time	two terms	train	45	10.70	5.05	35.95	75.68
Mac Studio M4 Max	absolute time	two terms	test	15	38.17	32.69	72.58	81.63
RTX 5080	absolute time	one term	train	36	14.18	11.77	28.14	45.36
RTX 5080	absolute time	one term	test	6	44.01	31.40	97.73	105.71
RTX 5080	absolute time	two terms	train	36	12.70	9.51	30.29	40.59
RTX 5080	absolute time	two terms	test	6	26.08	24.58	51.38	51.91
all hosts	absolute time	one term	train	126	13.94	8.17	34.00	96.54
all hosts	absolute time	one term	test	33	24.48	15.60	60.69	105.71
all hosts	absolute time	two terms	train	126	11.30	5.53	33.61	77.90
all hosts	absolute time	two terms	test	33	36.11	32.69	71.30	82.26
MacBook Pro M4 Max	relative time	one term	train	45	11.40	4.35	28.13	83.22
MacBook Pro M4 Max	relative time	one term	test	12	15.17	11.87	40.27	53.89
MacBook Pro M4 Max	relative time	two terms	train	45	8.53	5.06	18.93	44.61
MacBook Pro M4 Max	relative time	two terms	test	12	22.20	20.37	40.79	48.48
Mac Studio M4 Max	relative time	one term	train	45	14.24	6.99	32.74	88.15
Mac Studio M4 Max	relative time	one term	test	15	17.47	14.72	41.28	61.04
Mac Studio M4 Max	relative time	two terms	train	45	9.62	4.87	20.97	49.20
Mac Studio M4 Max	relative time	two terms	test	15	24.98	24.90	42.73	48.71
RTX 5080	relative time	one term	train	36	12.55	11.72	22.83	34.79
RTX 5080	relative time	one term	test	6	40.81	29.12	83.35	90.75
RTX 5080	relative time	two terms	train	36	11.74	10.36	21.21	28.73
RTX 5080	relative time	two terms	test	6	27.64	23.25	52.96	55.20
all hosts	relative time	one term	train	126	12.75	8.34	28.06	88.15
all hosts	relative time	one term	test	33	20.88	12.31	53.36	90.75
all hosts	relative time	two terms	train	126	9.84	5.95	21.21	49.20
all hosts	relative time	two terms	test	33	24.45	23.26	47.47	55.20

8 Provenance and explicit limits

The on-disk inputs used for this appendix have the following SHA-256 digests. The repository HEAD at generation was 28358bc864720887cf9b6e14aebf8e1e3ff86f55; the generator does not assert a clean working tree, so the file digests, rather than this commit alone, identify the inputs.

Table 6: Exact inputs used to generate this appendix.

Role	Path	SHA-256
measurements (lun-mac)	results/measurements_lun-mac.csv	99dcbf9abb9454877713198909703c1f77e00555f0258f7bf9d83e4395be99b4
calibration (lun-mac)	results/calibration_lun-mac.json	7c374f217b54540deb3266af88ba4ba393bfb85ac94e87147a9c32ffde0696bc
environment (lun-mac)	results/env_lun-mac.json	10ae655545911c2b668dcd8b629effa2fd369170b14dbd38f1451d6094f6f4a6
measurements (mac-studio-m4-max)	results/measurements_ mac-studio-m4-max.csv	a9a778237b2cc3cc297dd5940a930ece0cabd5445c1665b3de4c024a336bacf8
calibration (mac-studio-m4-max)	results/calibration_mac-studio-m4-max. json	033df8143068779f7c6726e3bdf7b5d9f77a8fcab3081b9200fd422ba3147fa
environment (mac-studio-m4-max)	results/env_mac-studio-m4-max.json	ebeee6b07a509c0ebba2dbd7938b9e585496e65a35a76a50d4514be5c75ec9c8
system profile (mac-studio-m4-max)	results/system_profile_ mac-studio-m4-max.txt	e1ba22cb0ef3d3b7ce48fa1337e0c02ef9d7193ade113afabe997c799d51ae88
operating system (mac-studio-m4-max)	results/os_mac-studio-m4-max.txt	cd5d6399b3785b04ceaae75045a9ca9f80f0bbc3526eff9d0c36b170ed1ec348
source commit (mac-studio-m4-max)	results/source_commit_ mac-studio-m4-max.txt	f457f4d50e3197d511eefc85b20fcfe11a8b8c250924b4c4d89b56b4ba8bc6df
measurement-source SHA-256 (mac-studio-m4-max)	results/measurement_source_tree_ mac-studio-m4-max.sha256	7c15bd94f60ebdf349cce6df465576fb4efb9dff098e8ffe4121ea72ed34dab0
llama.cpp package (mac-studio-m4-max)	results/llama_cpp_package_ mac-studio-m4-max.txt	61c81bc76cf5830ec3477189ec72d3d2b48a76ea2f52d0fa2531ad3b2cd6b91e
llama-bench SHA-256 (mac-studio-m4-max)	results/llama_bench_mac-studio-m4-max. sha256	6bf25f6aa7546aa438b8d82d96f6dcf02579ef22ceb14dac8e5a0d666f04fe22
model sources (mac-studio-m4-max)	results/model_sources_ mac-studio-m4-max.json	aed4539909710d8d145fd7d3b42fcc37047413cc502abd77d78b8edbb10be342
model integrity (mac-studio-m4-max)	results/model_integrity_ mac-studio-m4-max.json	ab8a4993e5a3f341cec3712431dc9bd351968c9d92aebf2b0fd95aefedaf7f4d
Python packages (mac-studio-m4-max)	results/python_packages_ mac-studio-m4-max.txt	480c7f9805b2d84e885c8b9b720cf538d0044990f04f76ab285b717c98f478f5
Tectonic package (mac-studio-m4-max)	results/tectonic_package_ mac-studio-m4-max.txt	e99453250832d16d9a603ea0837f7fc3f7bcb5402927bbf7edb7048a15953e0b
Tectonic SHA-256 (mac-studio-m4-max)	results/tectonic_mac-studio-m4-max. sha256	a6c2f0af071c38fc10f8b9ca56372535e4b63be5b36d76f533bd6a9cba8c9248
measurements (rtx5080)	results/measurements_rtx5080.csv	d4373736f7a38bb9c80eeec7804681af073e01d78ca9ff0b37e80b6f442deca4
calibration (rtx5080)	results/calibration_rtx5080.json	c8dc32dc71f547c9a3f2afc742f4342b9b229b6cb14665419196a57dacb0032f
environment (rtx5080)	results/env_rtx5080.json	bb4573ef46f13152005eab42226806824d0873dc0439fdb4d283b3811074fbaa

Continued on next page

Table 6: Exact inputs used to generate this appendix. (continued)

Role	Path	SHA-256
decode predictions	results/predictions.csv	6dd0b486fde80d561731b17976864280bb24111abea33f1b55ffedab6a863f79
prefill predictions	results/predictions_prefill.csv	9f401bc73378167c43a99caca58e89b7a4b10da0e5adfbfa4e31ae3316aee310
decode aggregate error table	results/error_table.csv	07b6e4847e4a3268ccbe632265bd8a4270c0d459432226e6a38b312572ef7612
decode error table by host	results/error_table_by_host.csv	f5a2338135fb06bf339ee5180639a34cc34a0e8643ae3f3c35993c4827ffa5e2
prefill aggregate error table	results/error_table_prefill.csv	460f1be7dfa5e64219f54405f8a40039dbad023800ab003ecef67dd53afb8786
prefill error table by host	results/error_table_prefill_by_host.csv	b5b0e18f1896fd1f1b4c4266a29b9a19f271ad5cc46bb32ea18bd43a30308194
manifest	results/model_manifest.json	2a672d622a30fd2107a01735744de9ba24e97025c3b9b973b5df442b2b662ec6
metadata	results/model_metadata.json	cb1afadb7fad3b6d9d2df88b7176d72f4c25068bbb865e8c9a33cc79a3e4a2e5
repeatability (lun-mac)	results/repeatability_lun-mac.csv	f79f9565aebc662bde98d1fef44b29d73e83188c60b6bf60a1061cc2de9f622
archived uncontrolled comparison	results/contaminated/measurements_UNCONTROLLED_lun-mac.csv.bak	b344118d3902a9aa6530a6113a3c5d0c0310ae2a703ad4986c8146d70821eb50
analysis implementation	llmperf/analyze.py	7eefe01436ae78f1ebe32c39c236a6f0dc521e35b410145ae785f04f66e8e27c
refinement implementation	llmperf/refine.py	d52acc54f91c253e1f05df31f5903e3c155584c0d1071643677ded87cf6b9e58
metadata implementation	llmperf/common.py	21f8da71de805e7eecbaa5dd9f5c8fc45f5fb2e90b67ce41bcfac8d3da18b973
publication-figure wrapper	paper/icassp2027/generate_main_figures.py	4c0b7ecaa86b523c70df76f3c79d943bb5dfb75aba0117c93d15433d7d390ee9
appendix generator	paper/icassp2027/generate_supplement.py	f557811bc7d7084fd7acba1c3d61c236b7e40ec1dc4d12757625c3da9817e934

- MacBook Pro M4 Max, Mac Studio M4 Max, and RTX 5080 contributed unequal, overlapping cohorts. Headline coefficients are fit separately by measured host. Each leave-one-host-out audit fits on all remaining measured hosts and does not establish universal transfer to new runtime stacks or formats.
- Every scored row records a requested `n_gpu_layers=99`, but no device-memory trace or runtime-reported resident-layer count was retained. This is a maximal-offload request, not proof of full accelerator residency; no partial-offload sweep or offload-cliff result exists.
- The largest RTX modeled file-plus-KV footprint is 17.508 GB for `Qwen3.8-27B-UD-Q3_K_XL.gguf` at depth 16,384, while calibration reports 17.094 GB of device memory. The differing accounting conventions further preclude a residency claim.
- Runtime provenance is host-specific and is reproduced in the environment and auxiliary provenance files. Placeholder commits, project commits, and usage-text version output are not interpreted as immutable runtime-binary revisions.
- The Studio acquisition records immutable repository revisions and per-file LFS SHA-256 identifiers for all 23 manifest entries. These identify the remote artifacts; they are not retrospective local-file hashes for the earlier MacBook and RTX copies.
- Derived parameter counts and per-depth KV bytes are consumed from the frozen metadata snapshot during appendix generation. This reproduces the analysis but is not a fresh, independent metadata extraction.
- Held-out coverage is small: 4 configurations on MacBook Pro M4 Max, 5 configurations on Mac Studio M4 Max, and 2 configurations on RTX 5080. Its observed quantization formats are MXFP4, Q4_K, Q4_K_M. No learning curve establishes how many reference configurations suffice.
- Decode MAPE treats three depths from each host–file configuration as observations; those rows are correlated. No narrow confidence claim is warranted.
- Prefill’s primary result is restricted to depth 0. Larger-depth predictions are scope diagnostics because the equation has no existing-prefix term. The retry rule is decode-only; the counts of prefill rows above each host’s declared CV gate are 7 of 57 on MacBook Pro M4 Max, 0 of 60 on Mac Studio M4 Max, and 29 of 42 on RTX 5080.
- Calibration-probe exclusions are host-specific (3 on MacBook Pro M4 Max, 3 on Mac Studio M4 Max, and 0 on RTX 5080). Failed configurations and probe identities are listed above; a recorded prompt-batch failure is not relabeled as an out-of-memory failure without supporting telemetry.
- Qwen3.8-27B IQ2 has 64 layers and 26.90B parameters, whereas the other shown Qwen3.8-27B files have 65 layers and 27.32B parameters. The quantization plot is therefore a near-family comparison, not a controlled bit-format substitution for one identical network.

The metadata snapshot additionally records these parser/source Git object IDs:

Object	Identifier
Code commit	1bdce5e0d917dce152cba4809d05fc1922a98797
Measurement blob	0ec72c61f2eaa371ba4f591a28964ec602d0686b
Predictions blob	de86a4286026e3b895b3c3809f9b4b0eb7dbda47
Shape-log blob	fa3d7b9cbf4a731219c7d52829fde031c67e3a2d